# Word Codes (Set B)

Year 4 Pt 2 · **Lesson Worksheet**

## Worked Examples

**Example 1 — Decoding with a Shift:** In a code, every letter is shifted forward by 2 (A→C, B→D...). What does the code word 'FQI' decode to?

**Answer:** DOG — shifting each letter BACK by 2: F→D, Q→O, I→G.

**Example 2 — Encoding with a Shift:** Using the same +2 shift code, encode the word 'CAT'.

**Answer:** EVC... check: C→E, A→C, T→V, so the code is 'ECV'. (Each letter moves forward 2 places: C→E, A→C, T→V.)

**Example 3 — Worked Pair Method:** If CAT is coded as DBU, what shift is being used, and how would you code the word DOG?

**Answer:** The shift is +1 (each letter moves forward one place: C→D, A→B, T→U). Applying +1 to DOG: D→E, O→P, G→H, giving 'EPH'.

## Spotting the Code (15 questions)

1. In a code, every letter is shifted forward by 1 (A→B, B→C...). What does "DPH" decode to?
2. Using a +1 shift code, encode the word "CAT".
3. If the code for DOG is EPH (shift +1), what is the code for PIG? A) QJH  B) QJI  C) RJH  D) QKH  E) PJH
4. Select ALL that apply: using a shift of +3 (A→D, B→E, C→F...), which of these are correctly encoded? A) CAT → FDW  B) DOG → GRJ  C) PIG → SLJ  D) SUN → VXQ  E) BAT → EDW
5. In a code, every letter is shifted BACK by 1 (B→A, C→B...). What does "EBU" decode to?
6. If BAT is coded as CBU (shift +1), what is the code for RUN? A) SVO  B) SVN  C) TVO  D) SUO  E) RVO
7. Using a +2 shift code (A→C, B→D...), encode the word "HEN".
8. In a code, every letter is shifted BACK by 2 (C→A, D→B...). What does "KQI" decode to?
9. If the word FOX is coded as GPY (shift +1), decode the word "JBU" using the same shift.
10. Select ALL that apply: using a shift of -1 (B→A, C→B...), which of these are correctly decoded? A) "DPH" → COG  B) "UBU" → TAT  C) "CBU" → BAT  D) "GPY" → FOX  E) "KQI" → LPH
11. Using a +1 shift code, encode the word "PIG".
12. If the word CAT is coded as BZS (shift -1), what is the code for BAT? A) AZS  B) BZS  C) AYS  D) ZYS  E) AZT
13. In a code, every letter is shifted forward by 2 (A→C, B→D...). What does the code word "LGZ" decode to?
14. Using a +3 shift code, encode the word "CAT".
15. Select ALL that apply: if the shift used is +2 (A→C, B→D...), which of these pairs is correctly matched? A) HEN→JGP  B) PIG→RKI  C) DOG→FQI  D) SUN→UWP  E) CAT→EDV

## Applying the Code (15 questions)

© PassFrame Tuition — for registered student use

1. In a code, CAT is written as ECV. Work out the shift being used, then use it to encode "DOG".
2. In a code, each letter's code is found by shifting forward by 4 places. Decode the word "GSR".
3. A code represents each letter by shifting back 3 places. If "PTFJ" is a coded word, decode it.
4. Select ALL that apply: a code shifts every letter forward by 5. Which of these pairs are correctly matched? A) DOG→ITL B) CAT→HFY C) PIG→UNL D) SUN→XZS E) FOX→KTC
5. In a code, the word BIRD is written as DKTF. Work out the shift, then encode the word "NEST" using the same shift.
6. A code shifts letters BACK by 4 places. Decode the coded word "PEJ".
7. If the word STAR is coded as VWDU (shift +3), decode the word "ZLQG" using the same shift.
8. In a code, each letter is replaced by the letter 5 places further on in the alphabet, wrapping round after Z if needed. Encode the word "WOLF".
9. Select ALL that apply: a code shifts every letter back by 6 places (wrapping round before A if needed). Which of these pairs are correctly matched? A) GOAT→AIUN B) MOON→GIIH C) HORSE→BILMY D) ZEBRA→TYVLU E) FROG→ZLIA
10. In a code, DUCK is written as FWEM. Work out the shift, then decode "JQWUG" using the same shift.
11. A code shifts every letter forward by 6 places. Encode the word "FISH".
12. If the word MOUSE is coded as PRXVH (shift +3), decode the word "ILVK" using the same shift.
13. In a code, each letter is replaced by the letter 7 places back in the alphabet, wrapping round before A if needed. Decode the coded word "UBKW".
14. Select ALL that apply: a code shifts every letter forward by 4 places. Which of these pairs are correctly matched? A) BEAR→FIEV B) SWAN→WAER C) GOAT→KSEX D) LION→PMSR E) DEER→HIIV
15. In a code, the word HORSE is written as KRUVH. Work out the shift, then use it to encode the word "SNAKE".

## Codes in Context (15 questions)

1. A code pair is given: TIGER → WLJHU. Work out the shift being used, then decode the word "VKDUN" using the same shift.
2. In a code, each letter's position number is DOUBLED, then converted back to a letter (wrapping round after Z, using the remainder when dividing by 26, with remainder 0 meaning Z). Encode the word "CAB" using this rule.
3. Select ALL that apply: a two-step code first shifts every letter forward by 3, then reverses the letter order of the whole word. Which of these results are correct for the word "DOG"? A) First step gives GRJ. B) Second step (reversing GRJ) gives JRG. C) The final coded word is JRG. D) The final coded word is GRJ. E) Reversing "DOG" first would give "GOD".
4. In a code, vowels (A, E, I, O, U) are shifted forward by 1, but consonants stay the same. Encode the word "CAT" using this rule.
5. A puzzle gives the pair LEMON → OHPRQ as a worked example. Work out the shift, then use it to decode "ULYHU".
6. In a code, each letter is replaced by the letter that is the same number of places from the END of the alphabet as it is from the START (so A ↔ Z, B ↔ Y, C ↔ X…). Encode the word "CAT" using this reversal rule.
7. Select ALL that apply: a code shifts consonants back by 2 places but leaves vowels unchanged. Applying this to the word "LEMON", which of these are correct? A) L becomes J. B) E stays E. C) M becomes K. D) O stays O. E) N becomes L.

© PassFrame Tuition — for registered student use

8. A code works in two steps: first replace each letter with its alphabet position number, then add 10 to each number (not converting back to letters). What is the resulting set of numbers for the word "CAB"?

9. In a code, the pair GRAPE → JUDSH is given as a worked example. Work out the shift, then encode the word "LEMON" using the same shift.

10. A two-step code shifts every letter back by 2, then swaps the first and last letters of the resulting word. Apply this to the word "HOUSE" and give the final result.

11. Select ALL that apply: a code reverses the alphabet (A ↔ Z, B ↔ Y, C ↔ X...). Applying this to the word "BAT", which of these are correct? A) B becomes Y. B) A becomes Z. C) T becomes G. D) The coded word is YZG. E) The coded word is ZYG.

12. In a code, each letter's position number is tripled, then the ones digit of the result is converted back to a letter (so a result of 23 would give the letter for 3, which is C; a result of 30 would give the letter for 0, treated as Z). What does the letter E become under this rule?

13. A puzzle gives the pair APPLE → DSSOH as a worked example. Work out the shift, then decode "VSULQJ" using the same shift.

14. In a code, each vowel is replaced by the NEXT vowel in the sequence A, E, I, O, U, A... (so A→E, E→I, I→O, O→U, U→A), while consonants stay the same. Encode the word "MOUSE".

15. A two-step code first reverses the letters of the word, then shifts every letter forward by 1. Apply this to the word "CAT" and give the final 3-letter result.

© PassFrame Tuition — for registered student use